Marcus Cedric S. Pedrosa

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Education

Polytechnic University of the Philippines — Sta. Mesa, Manila

Diploma in Computer Engineering Technology - Specializing Machine Learning — 2023 - 2026

Bachelor of Science in Computer Engineering — 2026 - Present

Relevant Coursework: Operating Systems, Software Development, Artificial Intelligence

Experience

Snapp Ventures Inc. — Alabang, National Capital Region

Software Quality Assurance Intern — March 2026 – May 2026

- Developed Playwright test scripts to automate regression testing and reduce repetitive manual QA.
- Designed SQL schemas across 5 relational tables in Directus and configured automated workflows to support HR leave request submission, approval, and status tracking.
- Validated REST API endpoints using Postman by analyzing API payloads and documenting backend issues.
- Executed end-to-end functional and security testing on Android APK mobile builds and staging sites, capturing multi-modal layout overlaps and securing role-based UI access permissions.

Projects

F1 2026 Winner Prediction System

- Developed a machine learning model that combines historical race data and the current season performance to predict the Formula 1 World Drivers' Champion using Monte Carlo simulation.
- Integrated a dynamic weightage system to account for the drastic 2026 technical regulation changes, shifting model reliance toward electrical powertrain efficiency and active aero adaptability.
- Built an ETL pipeline to ingest and transform telemetry, weather, and race data from the OpenF1 API.
- Designed a PostgreSQL database to store telemetry trends, race results, driver standings, and track-specific performance data for real-time model inference.

AI Bifrost Middleware

- Reduced redundant LLM calls and token usage through exact and semantic prompt caching.
- Improved response accuracy by routing time-sensitive queries through real-time web search with Tavily.
- Increased reliability with automatic Groq-to-Gemini failover during rate limits.

Budget Tracker Website

- Developed a full-stack budget tracker with AI receipt scanning, recurring expenses, and real-time budgeting.
- Implemented AI receipt scanning to extract up to 50 receipt items and automatically categorize expenses.
- Built spending analytics with 90-day anomaly detection, AI budget insights, and exportable financial reports.
- Implemented role-based authentication, with email verification, CSRF protection, and rate limiting.

Technical Skills

Frontend Development: HTML, Tailwind CSS, JavaScript, React, TypeScript

Backend & Database: Next.js, Python, Java, NumPy, Pandas, PostgreSQL, Flask, FastAPI, REST API, OpenAI/Gemini API

Tools: Git, GitHub, Docker, Postman, n8n, Resend, Claude Code, Vercel, Fly.io

Certifications

Computer Systems Servicing NC II — TESDA | September 2024

Introduction to Python Programming — Coursera | November 2025

Python Data Associate — DataCamp | February 2026